

## MEMMINGEN, Germany

WMO: 109470

Lat: 47.989N

Lon: 10.240E

Elev: 629

StdP: 93.99

Time Zone: 1.00 (EUC)

Period: 90-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.5      | -10.1 | -13.0                         | 1.3 | -12.1 | -11.1 | 1.6 | -9.6 | 13.6                  | 7.8  | 12.2 | 5.3  | 1.6                   | 60   | 0.563 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 11.3                   | 29.0            | 18.7 | 27.1 | 18.0 | 25.3 | 17.1 | 19.3                | 27.2 | 18.5 | 25.9 | 17.6 | 24.3 | 2.5                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.2                            | 12.5 | 22.6 | 15.6 | 11.9 | 21.7 | 14.8 | 11.4 | 20.9 | 57.7          | 27.4 | 54.7 | 26.0 | 52.1 | 24.4 | 21.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 10.9          | 9.1  | 7.7 | DB                         | -16.7 | 31.9               | 2.2 | 1.5                                                | -18.2 | 32.9       | -19.5 | 33.8       | -20.7 | 34.7       | -22.3 | 35.8 | (4) |
| (5)               |      |     | WB                         | -16.7 | 20.8               | 2.2 | 0.8                                                | -18.3 | 21.4       | -19.5 | 21.9       | -20.7 | 22.4       | -22.3 | 23.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.4         | -0.5  | 1.1   | 4.6   | 7.2   | 12.6  | 15.1  | 17.4  | 17.7  | 13.1  | 8.8   | 3.2   | 0.1   |  |  |
|                                                                        | DBStd       | 7.55        | 4.59  | 4.87  | 3.97  | 3.93  | 3.65  | 3.63  | 3.28  | 3.27  | 3.07  | 4.00  | 3.98  | 4.47  |  |  |
|                                                                        | HDD10.0     | 1459        | 327   | 249   | 172   | 103   | 17    | 2     | 0     | 1     | 7     | 69    | 206   | 308   |  |  |
|                                                                        | HDD18.3     | 3697        | 585   | 482   | 426   | 335   | 181   | 108   | 57    | 51    | 157   | 294   | 455   | 566   |  |  |
|                                                                        | CDD10.0     | 881         | 0     | 1     | 4     | 18    | 97    | 157   | 229   | 239   | 101   | 33    | 1     | 0     |  |  |
|                                                                        | CDD18.3     | 76          | 0     | 0     | 0     | 0     | 2     | 13    | 28    | 32    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 883         | 0     | 0     | 0     | 2     | 54    | 154   | 300   | 348   | 24    | 1     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 197         | 0     | 0     | 0     | 0     | 4     | 28    | 78    | 86    | 1     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.0         | 3.3   | 3.4   | 3.6   | 3.1   | 2.7   | 2.7   | 2.7   | 2.3   | 2.5   | 2.9   | 3.0   | 3.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1238        | 76    | 74    | 85    | 83    | 124   | 144   | 142   | 131   | 100   | 88    | 95    | 95    |  |  |
|                                                                        | PrecMax     | 1624        | 172   | 171   | 204   | 182   | 249   | 237   | 289   | 260   | 180   | 177   | 230   | 193   |  |  |
|                                                                        | PrecMin     | 992         | 4     | 29    | 28    | 14    | 15    | 42    | 25    | 54    | 45    | 20    | 8     | 14    |  |  |
|                                                                        | PrecStd     | 158         | 41    | 37    | 41    | 42    | 56    | 48    | 60    | 56    | 40    | 40    | 48    | 44    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 11.7  | 15.9  | 18.1  | 23.0  | 27.1  | 29.7  | 31.2  | 31.5  | 26.0  | 22.0  | 15.0  | 11.8  |  |  |
|                                                                        |             | MCWB        | 7.0   | 10.0  | 10.8  | 12.7  | 15.8  | 18.3  | 20.0  | 19.0  | 17.3  | 15.2  | 10.8  | 8.0   |  |  |
|                                                                        | 2%          | DB          | 9.1   | 12.6  | 15.5  | 20.1  | 24.9  | 27.0  | 29.0  | 29.0  | 23.7  | 19.2  | 11.9  | 9.2   |  |  |
|                                                                        |             | MCWB        | 6.1   | 7.6   | 9.6   | 12.2  | 15.4  | 17.8  | 18.9  | 18.7  | 16.3  | 14.0  | 8.7   | 6.6   |  |  |
|                                                                        | 5%          | DB          | 7.2   | 9.8   | 13.1  | 17.2  | 22.7  | 25.0  | 26.7  | 27.0  | 21.7  | 16.8  | 10.0  | 7.8   |  |  |
|                                                                        |             | MCWB        | 4.9   | 5.9   | 8.4   | 10.8  | 14.5  | 16.7  | 18.3  | 18.0  | 15.6  | 12.9  | 7.6   | 5.8   |  |  |
|                                                                        | 10%         | DB          | 5.8   | 7.5   | 10.9  | 14.7  | 20.4  | 22.8  | 24.5  | 25.1  | 19.2  | 14.6  | 8.8   | 6.3   |  |  |
|                                                                        |             | MCWB        | 3.8   | 4.8   | 7.2   | 9.3   | 13.4  | 15.8  | 17.2  | 17.2  | 14.3  | 11.7  | 7.0   | 4.7   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.4   | 9.6   | 11.7  | 13.7  | 17.1  | 19.7  | 20.4  | 20.1  | 18.1  | 16.3  | 11.0  | 8.7   |  |  |
|                                                                        |             | MCDB        | 10.6  | 14.8  | 17.4  | 20.6  | 24.6  | 26.6  | 29.3  | 28.2  | 24.8  | 20.7  | 14.3  | 11.2  |  |  |
|                                                                        | 2%          | WB          | 6.3   | 7.8   | 10.0  | 12.5  | 15.9  | 18.3  | 19.4  | 19.2  | 16.8  | 14.5  | 9.1   | 6.8   |  |  |
|                                                                        |             | MCDB        | 8.6   | 11.8  | 14.5  | 19.1  | 23.7  | 25.6  | 27.5  | 27.3  | 22.4  | 18.1  | 10.9  | 8.6   |  |  |
|                                                                        | 5%          | WB          | 5.1   | 6.2   | 8.8   | 11.1  | 14.9  | 17.2  | 18.5  | 18.4  | 15.9  | 13.1  | 8.1   | 5.9   |  |  |
|                                                                        |             | MCDB        | 7.1   | 9.4   | 12.5  | 17.1  | 21.8  | 24.2  | 25.9  | 26.0  | 20.9  | 16.1  | 10.0  | 7.5   |  |  |
|                                                                        | 10%         | WB          | 3.8   | 4.8   | 7.5   | 9.6   | 13.7  | 16.1  | 17.6  | 17.6  | 14.6  | 12.0  | 6.9   | 4.8   |  |  |
|                                                                        |             | MCDB        | 5.3   | 7.1   | 10.5  | 14.0  | 19.6  | 22.2  | 23.7  | 24.3  | 18.5  | 14.6  | 8.3   | 6.2   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.5   | 8.4   | 8.5   | 9.6   | 10.9  | 10.4  | 10.5  | 11.3  | 9.8   | 8.2   | 6.4   | 5.6   |  |  |
|                                                                        |             | MCDBR       | 9.0   | 12.5  | 12.7  | 14.7  | 15.4  | 14.7  | 14.7  | 15.1  | 14.1  | 12.1  | 9.6   | 7.5   |  |  |
|                                                                        | 5% WB       | MCWBR       | 6.5   | 8.3   | 7.9   | 8.5   | 7.9   | 7.3   | 7.0   | 7.0   | 8.0   | 7.7   | 6.7   | 5.5   |  |  |
|                                                                        |             | MCWBR       | 8.3   | 11.1  | 11.6  | 13.7  | 14.3  | 13.4  | 13.2  | 13.6  | 12.5  | 11.4  | 8.3   | 6.9   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.287 | 0.313 | 0.348 | 0.378 | 0.389 | 0.395 | 0.396 | 0.390 | 0.365 | 0.339 | 0.310 | 0.284 |  |  |
|                                                                        |             | taud        | 2.442 | 2.372 | 2.343 | 2.291 | 2.300 | 2.315 | 2.302 | 2.337 | 2.386 | 2.448 | 2.478 | 2.453 |  |  |
|                                                                        | Ebn at Noon |             | 797   | 846   | 867   | 872   | 874   | 868   | 861   | 850   | 839   | 806   | 761   | 755   |  |  |
|                                                                        |             | Edh at Noon | 65    | 86    | 104   | 121   | 125   | 124   | 124   | 115   | 99    | 80    | 63    | 57    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.12  | 1.99  | 3.12  | 4.28  | 4.93  | 5.56  | 5.37  | 4.72  | 3.46  | 2.23  | 1.22  | 0.92  |  |  |
|                                                                        | RadStd      |             | 0.14  | 0.27  | 0.39  | 0.58  | 0.55  | 0.51  | 0.49  | 0.48  | 0.41  | 0.25  | 0.16  | 0.13  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |  |  |
| (47) Regional (92 neighbors) |  | N/A   | N/A     | N/A    | N/A     | +0.42 | +0.40 | N/A         | N/A     | +55     | +15     |  |  |

Nomenclature: See separate page